bloodo

Laboratory: LifeLab1
Address: Dariaus ir Girėno g. 32A-1, LT-02188 Vilnius
e-mail: info@lifelab1.com
License No.: 4864

Protocol generation date: 2026-10-13
No.: 261006004

LABORATORY TEST PROTOCOL

Patient code: DBUQEWDT2449
Age: 59
Sex: Male

Sample type: dried blood spot (DBS)
Sample collected: 2026-09-23
Sample received: 2026-10-06

Test	Analysis date	Result	Measurement units	Interpretation	Method	Employee license No.
Nicotinamide adenine dinucleotide (NAD)	2026-10-13	14.1	µmol/L	Insufficient NAD level	2	MB000622

Analysis method:

2 - Enzymatic cyclic assay, test code: XLT01690-0

Your total NAD (NAD+ and NADH) in capillary blood (µmol/L) is in YELLOW zone*:



Result Interpretations

Undersanding Insuficient NAD Levels

Insufficient NAD levels suggest that your cells may not be producing or recycling enough NAD to fully support optimal energy metabolism and cellular repair. This may not indicate a severe imbalance, but it's a sign that your body could benefit from additional support.

Simple lifestyle improvements - such as regular moderate exercise, nutrient-rich meals, quality sleep, and effective stress management - can help restore healthy NAD function. Supporting your body with vitamins and nutrients involved in NAD synthesis, such as niacin (vitamin B3), tryptophan, and certain amino acids, may also be beneficial.

Insufficient NAD levels can arise from temporary factors, including stress, infection, intense physical activity, poor nutrition, or irregular sleep patterns. If levels remain low despite a healthy lifestyle, it may be useful to discuss targeted supplementation or further testing with a healthcare professional.

For women, mild NAD fluctuations may also occur throughout the menstrual cycle. To ensure accurate monitoring, consider re-testing on the same cycle day each time.

*The reference range is personalized according to your age and sex. These values were established through a large population study involving 1,068 participants.
The final reference range was established after applying strict health-based exclusion criteria, using data obtained only from healthy individuals.

This test is not intended to diagnose, treat, cure, or prevent any disease or replace the medical advice and/or treatment obtained from a qualified healthcare practitioner.

Let's explore the context behind your results

NAD is your body's energy helper

NAD (short for nicotinamide adenine dinucleotide) is a naturally occurring molecule found in almost every cell of your body. It fuels your cells with energy and keeps your mitochondria - your body's "power plants" - running smoothly.

When your NAD levels are healthy, you tend to feel more energetic, focused, and physically strong. New research suggests that NAD supports metabolism, aids recovery after illness, and helps your cells stay strong and healthy as you age [1,2].

NAD is essential for DNA repair

External factors such as pollution, ultraviolet (UV) radiation, tobacco smoke, chemicals/toxins, and pathogens (like certain viruses and bacteria) can directly damage DNA or increase oxidative stress, leading to DNA mutations or strand breaks [3].

Internal factors such as replication errors, oxidative stress from metabolism, inflammation, and hormonal or immune imbalances also contribute to continuous DNA damage [4].

The body has complex DNA repair mechanisms, in which NAD^+ dependent enzymes like PARPs and sirtuins are crucial [5], to constantly fix this damage and maintain genomic stability.

When your body is under stress from things like lack of sleep, pollution or illness, it experiences more oxidative stress, which can lower your NAD levels. Low NAD levels mean your cells have less energy to repair DNA and protect themselves.

Checking your NAD levels and especially tracking how they change over time can give you valuable insights into your overall health, energy, and how well your body is recovering and repairing itself.

Why we want to keep NAD in the balance?

Maintaining your NAD levels within a healthy range is like giving your body a powerful tool for overall wellbeing.



Optimal NAD levels have been associated with:

- Improved mitochondrial function [6],
- Enhanced healthspan, and
- A decreased risk of age-related diseases [7],
- Improved blood plasma lipid profiles [8],
- Improved bone density [9], and
- Reduced oxidative stress [10].

On the flip side, insufficient NAD can lead to various challenges, such as

- Accelerated aging due to DNA damage,
- Fatigue and brain-fog,
- Poor sleep quality, and
- A decrease in physical and mental activity [11],
- Impaired metabolism
- Higher vulnerability to pathological changes in the central nervous system – low NAD is associated with diseases like Alzheimer's and Parkinson's [12]

Studies indicate that NAD levels tend to decrease due to increased NAD consumption in the body, driven by higher oxidative stress, greater DNA repair demands, various diseases, and other physiological or environmental factors - rather than age alone.

Recommendations Proven by Science

If your results indicate lower-than-optimal NAD levels, here are some research-based recommendations shown to help raise NAD levels. Remember, you are unique, and there is no single "right way" to increase your NAD levels. **The best results come from finding habits that work for you, staying consistent, and adjusting over time as your body adapts.**



Supplementation

NAD levels can be enhanced with NAD precursors such as nicotinamide riboside (NR) or nicotinamide mononucleotide (NMN). These precursors are converted in the body into NAD [13]. Please note that that supplementation or intravenous interventions should be approached carefully and under professional supervision. Individual response to the supplementation has to be evaluated and re-tested after use.



Regular exercise

Moderate to high endurance or resistance training can potentially increase NAD levels over time, particularly in muscle tissue. This occurs as the body adapts by enhancing its biological capacity to meet future physical demands.

It's important to note that while regular training supports NAD production, a single intense workout may temporarily deplete NAD due to high energy requirements [14]. Therefore, it's best to collect your blood samples for NAD test when you're well-rested and not overtrained, to ensure accurate results.



Fasting

Caloric restriction increases NAD levels while lowering NADH levels. Reduced NADH levels lead to increased activity of NAD - consuming enzymes, known as sirtuins [15]. Sirtuins are responsible for DNA repair, cell metabolism, and the stress response. Fasting can be used to boost NAD levels [16]. However, these approaches may not be suitable for everyone. Prolonged fasting or excessive calorie restriction can have adverse effects and may actually lower NAD levels.



Nutrition

A diet high in fatty or sugary foods can cause energy overload and decrease NAD levels. It also leads to elevated blood sugar and insulin levels, as well as increased oxidative stress and cellular damage [15]. Try to limit your intake of sugar and unhealthy fats, and include more vegetables such as edamame, broccoli, cucumber, avocado, and tomatoes in your diet [17].



Sleep quality

Maintaining a consistent sleep routine is one of the effective ways to support natural NAD production. Stable circadian rhythms promote balanced secretion of melatonin and cortisol, which are tightly linked to cellular energy metabolism and NAD synthesis [18,19]. Creating a dark, quiet, and cool sleep environment (17–28 °C, 40–60 % humidity) optimizes these rhythms and enhances nighttime cellular repair [20]. Inadequate or excessive sleep, however, can accelerate biological aging and reduce NAD availability, highlighting the importance of 6-8 hours of restorative sleep per night [21].



Avoiding alcohol

Alcohol disrupts metabolic processes by inducing oxidative stress and depleting NAD. This depletion impairs energy production and promotes inflammation and metabolic dysfunction. It is strongly recommended to avoid alcohol to prevent further damage to cellular health [22].

When to Re-Measure NAD Levels

NAD levels in the blood can respond relatively quickly to external and lifestyle factors, such as diet, sleep, stress, and physical activity. To ensure the most accurate and comparable results, please collect your sample in the morning before eating (fasted state) and avoid intense exercise beforehand, as strenuous activity may temporarily affect NAD metabolism.

3-4 months To evaluate long-term changes, repeat testing is recommended after 3-4 months of implementing NAD-supporting lifestyle measures.

2 weeks The effect of supplementation may be noticeable within one to two weeks of consistent use, although individual responses can vary.

For women, NAD levels can fluctuate throughout the menstrual cycle, and this variation tends to be individual but consistent for each person across cycles. Our data suggest that women should collect samples on the same cycle day for the most reliable tracking of NAD trends over time.

Control your energy and longevity



References:

1. Braidy N, Berg J, Clement J, Khorshidi F, Poljak A, Jayasena T, Grant R & Sachdev P (2019) Role of Nicotinamide Adenine Dinucleotide and Related Precursors as Therapeutic Targets for Age-Related Degenerative Diseases: Rationale, Biochemistry, Pharmacokinetics, and Outcomes. Antioxid Redox Signal 30, 251–294.
2. Cantó C, Menzies KJ & Auwerx J (2015) NAD(+) Metabolism and the Control of Energy Homeostasis: A Balancing Act between Mitochondria and the Nucleus. Cell Metab 22, 31–53.
3. Kazensky L, Matković K, Gerić M, Žegura B, Pehnec G & Gajski G (2024) Impact of indoor air pollution on DNA damage and chromosome stability: a systematic review. Arch Toxicol 98, 2817–2841.
4. Yu Y, Cui Y, Niedernhofer LJ & Wang Y (2016) Occurrence, Biological Consequences, and Human Health Relevance of Oxidative Stress-Induced DNA Damage. Chem Res Toxicol 29, 2008–2039.
5. Saville KM, Clark J, Wilk A, Rogers GD, Andrews JF, Koczor CA & Sobol RW (2020) NAD(+)-mediated regulation of mammalian base excision repair. DNA Repair (Amst) 93, 102930.
6. Li W & Sauve AA (2015) NAD+ content and its role in mitochondria. Methods Mol Biol 1241, 39–48.
7. Wilson N, Kataura T, Korsgen ME, Sun C, Sarkar S & Korolchuk VI (2023) The autophagy-NAD axis in longevity and disease. Trends Cell Biol 33, 788–802.
8. Zhong O, Wang J, Tan Y, Lei X & Tang Z (2022) Effects of NAD+ precursor supplementation on glucose and lipid metabolism in humans: a meta-analysis. Nutr Metab (Lond) 19, 20.
9. Kim H-N, Ponte F, Warren A, Ring R, Iyer S, Han L & Almeida M (2021) A decrease in NAD(+) contributes to the loss of osteoprogenitors and bone mass with aging. NPJ aging Mech Dis 7, 8.
10. Zhu Y, Zhao K-K, Tong Y, Zhou Y-L, Wang Y-X, Zhao P-Q & Wang Z-Y (2016) Exogenous NAD(+) decreases oxidative stress and protects H2O2-treated RPE cells against necrotic death through the up-regulation of autophagy. Sci Rep 6, 26322.
11. Kim M, Seol J, Sato T, Fukamizu Y, Sakurai T & Okura T (2022) Effect of 12-Week Intake of Nicotinamide Mononucleotide on Sleep Quality, Fatigue, and Physical Performance in Older Japanese Adults: A Randomized, Double-Blind Placebo-Controlled Study. Nutrients 14.
12. Li F, Wu C & Wang G (2024) Targeting NAD Metabolism for the Therapy of Age-Related Neurodegenerative Diseases. Neurosci Bull 40, 218–240.
13. Li X, Yang H, Jin H, Turkez H, Ozturk G, Doganay HL, Zhang C, Nielsen J, Uhlén M, Borén J & Mardinoglu A (2023) The acute effect of different NAD+ precursors included in the combined metabolic activators. Free Radic Biol Med 205, 77–89.
14. Walzik D, Jonas W, Joisten N, Belen S, Wüst RCI, Guillemin G & Zimmer P (2023) Specific effects of exercise as NAD + - boosting strategy: Current knowledge and future perspectives. Acta Physiol, 1–15.
15. Poljsak B, Kovač V & Milisav I (2020) Healthy Lifestyle Recommendations: Do the Beneficial Effects Originate from NAD(+) Amount at the Cellular Level? Oxid Med Cell Longev 2020, 8819627.
16. Dongoran RA, Mardiana M, Huang C-Y & Situmorang JH (2024) Boosting NAD+ levels through fasting to aid in COVID-19 recovery. Front Immunol 15, 1319106.
17. Mills KF, Yoshida S, Stein LR, Grozio A, Kubota S, Sasaki Y, Redpath P, Migaud ME, Apte RS, Uchida K, Yoshino J & Imai S-I (2016) Long-Term Administration of Nicotinamide Mononucleotide Mitigates Age-Associated Physiological Decline in Mice. Cell Metab 24, 795–806.
18. Ungurianu A & Marina V (2025) Melatonin and Cortisol Suppression and Circadian Rhythm Disruption in Burnout Among Healthcare Professionals: A Systematic Review. Clin Pract 15.
19. Ramsey KM, Yoshino J, Brace CS, Abrassart D, Kobayashi Y, Marcheva B, Hong H-K, Chong JL, Buhr ED, Lee C, Takahashi JS, Imai S-I & Bass J (2009) Circadian clock feedback cycle through NAMPT-mediated NAD+ biosynthesis. Science 324, 651–654.
20. Caddick ZA, Gregory K, Arsintescu L & Flynn-Evans EE (2018) A review of the environmental parameters necessary for an optimal sleep environment. Build Environ 132, 11–20.
21. Wang M, Yang M, Liang S, Wang N, Wang Y, Sambou ML, Qin N, Zhu M, Wang C, Jiang Y & Dai J (2024) Association between sleep traits and biological aging risk: a Mendelian randomization study based on 157 227 cases and 179 332 controls. Sleep 47, 1–11.
22. Lee J, Lee J-Y & Kang H (2025) Excessive alcohol consumption: a driver of metabolic dysfunction and inflammation. Front Toxicol 7, 1670769.